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A multi-objective production scheduling model and dynamic dispatching rules for unrelated parallel machines with sequence-dependent set-up times

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ABSTRACT

This study presents a production scheduling for unrelated parallel machines with machine and job sequence-dependent setup times. The system performance measures to minimize include makespan, total tardiness, and number of tardy jobs. The aim of the study is to develop a solution methodology that can solve the problem for the large scale. First, the problem is formulated as a mixed-integer linear programming model. The augmented ε-constraint method is applied to find Pareto solutions for small problem instances. The purpose is to demonstrate that Pareto solutions, which balance the trade-offs among three measures of performance, can be found for these instances. Dispatching rule-based heuristics is developed to solve large problem instances. The heuristics feature three dispatching rules that are designed to handle the dependent setup time. In addition, these rules are combined into six variants using a time-based rule-switching mechanism. The heuristics are tested with 18 problem instances, containing 244 to 298 jobs, in two demand scenarios derived from the monthly demand data from an industrial user. Under each demand scenario, a set of heuristics that provides the best performance with respect to the three measures is identified. The heuristics include combinations of the shortest completion time and due date-based rules. Finally, a multi-criteria decision-making analysis is performed to determine the conditions specified by the weight given to each measure, with which one heuristic is preferred over the others.

1. Introduction

Effective production planning is critical for manufacturing systems, particularly when dealing with complex job sequencing and scheduling tasks. This paper addresses these challenges for production systems involving non-identical machines, known as unrelated parallel machines. The problem is motivated by a real problem at the largest steel pipe manufacturer in Thailand. This production system consists of multiple machines that differ in their capabilities and production rates, adding complexity to scheduling as they operate in parallel. In each period (month), there are jobs, i.e., production orders, to be processed with their own production quantity and due date. All jobs can be processed at the beginning of each period with no job priority. This implies that a job may have different processing times for different machines.

The complexity is further increased by machine setup times. Specifically, the setup time of a job depends on the machine to which it is assigned and its sequence. For instance, if two adjacent jobs are in different families, the setup time is longer than if they are in the same family but are different in dimension. That is, the setup time for a job depends not only on the machine, but also on the preceding job on that machine. This is known as the machine- and sequence-dependent setup time.

In each period (i.e., production month), jobs representing orders from customers are placed before the beginning of the period such that production scheduling is performed during the previous period. Consequently, the setup operation of the first scheduled job on each machine is performed before the beginning of the period. In other words, the setup time of the first job on each machine is set to zero for each production month.

This problem is also commonly found in many manufacturing industries, where there are several product families to be made on the same set of unrelated machines. In most cases, the primary objective is to determine a job schedule for each machine such that the jobs are completed as early as possible. This is usually measured by the makespan, that is, the completion time of the last scheduled job. Moreover, these jobs usually have different promised delivery dates. To measure how well the due dates are met, the tardiness of each job and number of tardy jobs are calculated. In practice, minimizing the makespan, total tardiness, and number of tardy jobs are conflicting

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objectives. Therefore, it is in the best interest of a planner to balance these performance measures, such that production efficiency is sustained without sacrificing customer satisfaction. Although the trade-offs among the three measures of performance are well known in the scheduling literature, no study has attempted to offer a solution that strikes a balance among these measures for the problem under study.

This study makes three major contributions to literature.

- First, we develop a mixed-integer linear programming (MILP) model for the unrelated parallel-machine scheduling problem with machine- and sequence-dependent setup times. The model is adapted from the study by Avalos-Rosales et al. [1] to include the total tardiness and total number of tardy jobs in addition to the original objective function of minimizing the makespan. Moreover, we implement the augmented ε-constraint (AUGMECON) method to find compromise solutions among the three measures. The application of the proposed model is demonstrated using small problem instances. This contribution fills a gap in the literature, where existing models either simplify setup times or focus on single-objective optimization for unrelated parallel machines.
- Second, we design dynamic dispatching rule-based heuristics (DDR) for large problem instances. Specifically, three widely used single dispatching rules, including shortest processing time (SPT), earliest due date (EDD), and longest processing time (LPT), are modified to consider machine- and sequence-dependent setup times. This modification results in three dispatching rules designed for the sequence-dependent setup scheduling problems. In addition, the three rules are combined into six dispatching rules that feature rule switching times. The mechanism that allows the change of dispatching rules during the scheduling process is very effective not only in this study, but also in scheduling problems in general due especially to its ease of implementation.
- Third, based on the set of effective rules, a thorough multi-criteria decision-making analysis is performed to further identify the conditions in which each rule is most effective. These conditions are specified by the relative weights of importance that a decision-maker assigns to the three performance measures. The approach is believed to be the first of its kind that has been performed for a multi-objective scheduling problem. The benefit of this method can be explored in other scheduling problem settings with multiple objectives.

The remaining part of this paper is organized as follows. Relevant literature, research gaps, and contributions of this research are discussed in Section 2. The mathematical model is formulated in Section 3, while the heuristics are constructed in Section 4. The computational study and its results are discussed in Section 5. Conclusions of the research is summarized in Section 6.

2. Literature review

Production scheduling problems on parallel machines have been explored in numerous studies. These problems can be categorized using various criteria, such as parallel machine classification, setup time, system measures of performance, and solution methods. Parallel machines can be classified as identical, uniform, and unrelated. Identical parallel machines imply that the processing time of a job is the same for every machine, which has the same production rate [2–9]. Studies on uniform machines with different processing times and production rates can be found in Li et al. [10] and Zandi et al. [11]. Unrelated parallel machines have different production rates and specific machine restrictions, which means that each job can be processed by some machines [1,12–21].

Setup time is the time required to prepare resources to process the next task [22]. It is classified as sequence-independent, sequence-dependent, and machine-dependent. Sequence-independent setup time means that the setup time is constant regardless of the job or machine [18,23,24]. The sequence-dependent setup time depends on the preceding job on the machine [12,19,25–30]. In other words, sequence-dependent setup times are determined by a pair of adjacent jobs on a machine. The machine-dependent setup time depends on the machine assigned to process a job [1,26,28,30,31].

Several criteria are used for measuring the performance of scheduling solutions, including the makespan, total tardiness, and number of tardy jobs [32,33]. Studies focusing on minimizing the makespan have been conducted by Avalos-Rosales et al. [1], Arroyo et al. [20], Soper [21], Ezugwu [31], Shchepin and Vakhania [34], and Lee et al. [35]. Research on minimizing total tardiness includes works by Logendran et al. [12] and Yin et al. [36]. Finally, Su et al. [18] minimize the number of tardy jobs.

In addition to single-objective optimization, seminal contributions that take different objectives into account have emerged over the years in the field of unrelated parallel machine scheduling. Chayu and Chang [25] investigate the minimization of total weighted flow time and total weighted tardiness, dealing with job sequence- and machine-dependent setup times. Lin et al. [13] expand this research by targeting critical scheduling objectives, including makespan, total weighted completion time, and total weighted tardiness. Torabi et al. [26] explore the minimization of total weighted flow time, total weighted tardiness, and machine load variation. Nikabadi and Naderi [27] simultaneously minimize makespan, number of tardy jobs, earliness and tardiness. In a subsequent study, Wang and Alidaee [28] undertake the order acceptance and scheduling on unrelated parallel machines, introducing a multi-objective mixed-integer linear programming model that minimizes total workload and machine fixed costs. Farmand et al. [37] develop a bi-objective model that integrates identical parallel machines scheduling and supply chain management. The model aims at minimizing both time-based and cost-based measures of performance. The time-based measure consists of total weighted tardiness and operation time, while the cost-based contains penalty for tardy orders, earliness, and batch delivery cost. Lastly, Yepes-Borrero et al. [30] formulate a bi-objective model that minimizes makepsan and maximum number of resources needed.

For the methodology, a mixed-integer linear programming model (MILP) can solve only small problem instances. Therefore, many researchers have developed heuristics for solving large-scale problems. Popular heuristics for unrelated parallel machines include genetic algorithm [1,13,14,18], and simulated algorithm [17,18]. Other heuristics found in literature reviews include branch and bound [19], iterated greedy [20,30,38], polynomial-time approximation scheme [15], three-phase [29], and composite dispatching rule [19]. Yin et al. [36] modify the iterated greedy algorithm to their scheduling problem. In addition, Chyu and Chang [25] introduce a Pareto evolutionary approach for a two-objective (total weighted flow time and weighted tardiness) scheduling problem, involving job sequence- and machine-dependent setup times. Lin et al. [13] extend the work of Chyu and Chang [25] by modifying the genetic algorithm to find multi-objective (including makespan, weighted completion time, and weighted tardiness) solutions in scheduling problems. Their algorithm shows exceptional proficiency in generating non-dominated solutions. Torabi et al. [26] propose a multi-objective particle swarm optimization (MOPSO) algorithm. Nikabadi and Naderi [27] employ multi-objective genetic algorithm (MOGA) and simulated annealing (SA). Wang and Alidaee [28] develop a list-scheduling-based multi-objective parthenogenetic algorithm (LS-MPGA). Their computational experiments demonstrate the effectiveness of the LS-MPGA in achieving trade-off optimization objectives. Bektur and Sarac [29] propose two heuristics, simulated annealing and Tabu Search, which use an initial solution from modified apparent tardiness cost with setups (ATCS) dispatching rules. Farmand et al. [37] adopt meta-heuristic algorithms such as MOPSO and non-dominated sorting genetic algorithm II (NSGA-II) for solving an integrated problem of identical parallel machines scheduling and supply

Table 1
A summary of relevant research work.

Author	Problem characteristic		Measures of performance		Solution method
	Parallel machine	Setup time	Completion time	Tardiness	
Ghalami and Grosu [2]	I	N	M		PTAS
Kim and Lee [3]	I	J	M		SA-TL
Omar and Teo [4]	I	S		T, E	
Anderson et al. [5]	I	S		T, E	
Hamzadayi and Yildiz [6]	I	S	M		SA, GA
Ozer and Sarac [7]	I	S	WC		GA
Farmand et al. [37]	I	S		WT	MOPSO
Li et al. [10]	UF	N		T	SA
Zandi et al. [11]	UF	N	TC		PC
Lin et al. [23]	U	N	M, WC	WT	GA, HEU
Lin et al. [13]	U	N	M, WC	WT	GA
Ghirardi and Pott [14]	U	N	M		RBS
Lin and Ying [15]	U	N	M, WC	WT	MOMSA
Cheng and Huang [16]	U	N		T, E	GARTC
Arroyo et al. [20]	U	N	F		IG
Xiong et al. [17]	U	N	WC		BAP
Yin et al. [36]	U	N	TC		IG
Pei et al. [38]	U	N	WC		D-SDP, FD-SDP
Soper [21]	U	N	M		
Su et al. [18]	U	J		L	Three-phase
Logendran et al. [12]	U	S		WT	TS
Bajestani and Tavakkoli-Moghaddam [19]	U	S		WT	ATC, BAB
Bektur and Sarac [29]	U	S		WT	SA, TS, ATCS
Avalos-Rosales et al. [1]	U	MS	M		GA, MA
Ezugwu [31]	U	MS	M		HSOSSA
Shchepin and Vakhania [34]	U	N	M		Rounding approach
Chyu and Chang [25]	U	S	WF	WT	SA, GA
Nikabadi and Naderi [27]	U	S	M	T, E, L	SA, MOGA
Torabi et al. [26]	U	MS	WF	WT	MOPSO
Wang and Alidaee [28]	U	MS	M		LS-MPGA
Yepes-Borrero et al. [30]	U	MS	M		IG
This paper	U	MS	M	T, L	DDR

Note: I: Identical machine, UF: Uniform machine, U: Unrelated machine; N: Not consider, J: Job-dependent setup, S: Sequence-dependent setup, MS: Machine- and sequence-dependent; M: Makespan, TC: Total completion time, WC: Weighted completion time, F: Average flow time; T: Total tardiness, E: Total earliness, WF: Weighted flow time WT: Weighted tardiness, L: Number of tardy jobs, PTAS: Polynomial-time approximation scheme, SA: Simulated annealing, GA: Genetic algorithms, TS: Tabu search, RBS: Recovering beam search, MOGA: Multi-objective genetic algorithm, MOMSA: Multi-objective multi-point simulated annealing, GARTC: Genetic algorithms time control, IG: Iterated greedy, ATCS: Apparent tardiness cost with setups, HSOSSA: Hybrid symbiotic organisms search with simulated annealing, MOPSO: Multi-objective particle swarm optimization, LS-MPGA: List-scheduling-based multi-objective parthenogenetic algorithm, DDR: Dynamic dispatching rule-based heuristics.

chain management. Krim et al. [39] propose two novel metaheuristics based on tabu search, which leverages distinct neighborhood structures and solution spaces.

An overview of relevant literature on parallel machine scheduling is presented in Table 1.

According to Table 1, no research has studied a scheduling problem with multiple objective functions of makespan, total tardiness, and the number of tardy jobs in production systems of unrelated parallel machines with machine- and sequence-dependent setup times. Motivated by this research gap, this study aims to address the following aspects.

- The formulation of a multi-objective MILP model that can provide Pareto solutions, which represents the best trade-offs among makespan, total tardiness, and number of tardy jobs.
- The capability and limit of the MILP model is examined through a numerical experiment that is adapted from a practical problem.
- In addition to mathematical model, different dynamic dispatching rule-based heuristics are proposed, constructed, and experimented. The purpose is to offer an alternative approach that can find high-quality solutions within a reasonable time. This aligns with the need of industries, especially those with unrelated parallel machine systems, for a scheduling method that consumes a minimal amount of time for generating good production schedules.
- To the best of our knowledge, there is one previous study by Bektur and Sarac [29] that includes sequence-dependent setup time in the ATCS dispatching rule. Instead of constructing only one dispatching rule, our work modifies three different rules that are commonly employed in practice.

3. Model development

3.1. Problem statement

This study considers an unrelated parallel-machine scheduling problem with machine- and sequence-dependent setup times. The problem descriptions are as follows.

- There are n jobs to be processed in each period (i.e., a month). Each job consists of the required production quantity and the due date.
- All jobs are released for production at the beginning of each period, that is, all jobs have the same release time. There is no priority among jobs in terms of their importance.
- The production system consists of m unrelated parallel machines.
- The processing time of a job on a machine depends on the job-machine pair.
- Each machine can process a subset of all jobs. For the jobs that a machine cannot process, their processing times are set to a large value.
- Setup time is machine- and sequence-dependent, i.e., setup time of a job is determined by a pair of preceding job and the job itself on a given machine.

3.2. Mathematical models

The problem is formulated as a MILP model, where each model minimizes one of the three system performance measures, including the makespan, total tardiness, and number of tardy jobs. MILP models are used to study the characteristics of the problems with respect to different system measures. This is performed by solving the optimization problem using four small problem instances. In addition, another model that sought a set of compromise solutions among the three measures is formulated. The trade-off between the three measures is examined to identify the Pareto solutions. The model indices, parameters, and decision variables are as follows:

Index

i, j	indices of jobs, where N denotes the set of jobs, $i, j \in N = \{1, 2, \ldots, n\}$, and N_0 denotes the set of jobs including a dummy job 0, $N_0 = \{0\} \cup N$.
k	index of machines, where M denotes the set of machines; $k \in M = \{1, 2, \ldots, m\}$. Also, because each machine can process a different set of jobs, each given job j can only be processed by a subset of M, i.e., M_j. That is, $M = M_1 \cup M_2 \ldots \cup M_n$

Parameter

$P_{j,k}$	=	the processing time of job j on machine k
$S_{i,j,k}$	=	the required setup time of machine k, when the machine is set up to process job j and has a previous job i.
D_j	=	the due date of the job j
V	=	a large value
$NP_{j,k}$	=	the restriction condition of job j that indicates if it can be processed on machine k if it equals 1, 0 otherwise.

Decision Variable

$X_{i,j,k}$	=	a binary variable that takes on a value of 1, if job j is scheduled after job i on machine k, or 0 otherwise.
C_j	=	the completion time of job j (hours)
C_{max}	=	the makespan (the maximum completion time of all jobs, $j \in N$) (hours)
e_j^+	=	the tardiness of job j (hours)
e_j^-	=	the earliness of job j (hours)
U_j	=	a binary variable that is equal to 1 if job j is tardy, or 0 otherwise.

3.2.1. Minimize makespan

The MILP is adopted by Avalos-Rosales et al. [1] and Kongsri and Buddhakulsomsiri [40]. It is regarded as M1.

$$\text{Minimize } C_{max} \tag{1}$$

Subject to

$$\sum_{k \in M} \sum_{i \in N_0 : i \neq j} X_{i,j,k} = 1, \qquad \forall j \in N \tag{2}$$

$$\sum_{k \in M} \sum_{j \in N_0 : i \neq j} X_{i,j,k} = 1, \qquad \forall i \in N \tag{3}$$

$$\sum_{j \in N_0 : i \neq j} X_{i,j,k} - \sum_{h \in N_0 : h \neq i} X_{h,i,k} = 0, \qquad \forall k \in M, \forall i \in N \tag{4}$$

$$\sum_{j \in N} X_{0,j,k} \leq 1, \qquad \forall k \in M \tag{5}$$

$$C_j - C_i + V\left(1 - X_{i,j,k}\right) \geq S_{i,j,k} + P_{j,k}, \qquad \forall i \in N_0, \forall j \in N : i \neq j, \forall k \in M \tag{6}$$

$$C_0 = 0 \tag{7}$$

$$C_j \leq C_{max}, \qquad \forall j \in N \tag{8}$$

$$\sum_{k \in M} \sum_{i \in N_0 : i \neq j} X_{i,j,k} \leq NP_{j,k} \qquad \forall j \in N \tag{9}$$

$$C_j \geq 0, \qquad \forall j \in N \tag{10}$$

$$X_{i,j,k} \in \{0, 1\}, \qquad \forall i \in N_0, \forall j \in N_0 : i \neq j, \forall k \in M \tag{11}$$

In the model, the objective function (1) minimizes the makespan. Constraints (2) and (3) ensure that each job has only one preceding job and exactly one successive job. Constraint (4) represents the flow balance of each job on each machine. Constraint (5) specifies that each machine should start with a dummy job. Constraint (6) calculates each job completion time from the completion time of the preceding job, setup time, and job-processing time. Constraint (7) indicates that the dummy job had a completion time of 0. Constraint (8) determines the makespan from the completion time of all jobs. Constraint (9) determines that job j cannot be processed on machine m if it has a restriction. Finally, Constraints (10)–(11) identify the types of decision variables.

3.2.2. Minimize the total tardiness

To generate a schedule of jobs with minimum total tardiness, M1 is modified into another model, denoted M2, by introducing additional decision variables and constraints. Specifically, two variables, i.e. e_j^+ and e_j^- which represent the tardiness and earliness of job j, are included in M2. The objective function for minimizing the total tardiness is expressed by Eq. (12). In addition, the values and types of e_j^+ and e_j^- are specified by constraints (13) and (14), respectively.

$$\text{Minimize } T = \sum_{j \in N} e_j^+ \tag{12}$$

Subject to

(2)–(10)

$$C_j - D_j = e_j^+ - e_j^-, \qquad \forall j \in N \tag{13}$$

$$e_j^+, e_j^- \geq 0, \qquad \forall j \in N \tag{14}$$

3.2.3. Minimize the number of tardy jobs

Similar to M2, the model in this section, regarded as M3, is derived from M1 by adding a binary decision variable U_j to indicate whether job j is late. M3 is formulated as follows:

$$\text{Minimize } L = \sum_{j \in N} U_j \tag{15}$$

Subject to

$$e_j^+ = V \times U_j, \qquad \forall j \in N \tag{16}$$

$$U_j = \{0, 1\}, \qquad \forall j \in N \tag{17}$$

The objective function (15) minimizes the number of tardy jobs. The additional constraints (16) and (17) specify the tardy status of each job and the type of additional decision variable, respectively.

3.2.4. Determine compromise solutions among the three performance measures

To find compromise solutions, the multiple-objective model (M4) is formulated as follows:

Min $f_1 = C_{max}$
Min $f_2 = T$
Min $f_3 = L$
Subject to:

Constraints (2)–(11), (13)–(14), and (16)–(17)

Since multi-objective model involves optimizing several objective functions simultaneously, achieving a single solution that is optimal for all objectives that are in conflict is generally impossible. As a result, a widely adopted approach for dealing with these problems is the ε-constraint method. This technique facilitates the exploration of interactions among objectives such that a balanced compromise among them can be achieved [41]. In this study, we utilize an augmented version of the ε-constraint (AUGMECON) method [42] to address the proposed multi-objective model for production scheduling problem. The AUGMECON method is designed to generate efficient Pareto-optimal, non-dominated solutions in multi-objective problems, with proven effectiveness in producing pareto optimal results [43]. For a detailed explanation of this method, audiences are referred to Mavrotas [42].

The AUGMECON method is applied to M4 using the following steps:

Step 1: Set the objective functions of f_2 and f_3 as Constraints (18) and (19)

$$\sum_{j\in N} e_j^+ \le T \tag{18}$$

$$\sum_{j\in N} U_j \le L \tag{19}$$

where T denotes the specified acceptable total tardiness, and L denotes the specified acceptable number of tardy jobs, and are additional parameters of M4.

Step 2: Create the payoff table by solving M1, M2, and M3 to obtain the best and worst possible values of C_{max}, T, and L.

Step 3: Calculate the range of f_2 and f_3. Each range contained several grid points.

Step 4: M4 is solved for each combination of grid points in the ranges of T and L.

Step 5: Determine the Pareto solutions, which are the non-dominant solutions with respect to the three objective functions.

4. Dynamic dispatching rule based heuristics

Because only small problem instances can be solved by the MILP models presented in the previous section, heuristics, which implement dynamic dispatching rules, are developed in this section to solve large-scale problem instances that properly represent real problems. Our proposed heuristics utilize widely adopted dispatching rules, including the shortest processing time (SPT), longest processing time (LPT), and earliest due date (EDD). These rules are modified to consider the characteristics of our problem, that is, sequence- and machine-dependent setup-time setup times and unrelated parallel machines. The modified rules are denoted as SCT, SC-LPT, and SC-EDD, respectively. Their descriptions are as follows:

Notation:

N_i	=	a set of scheduled jobs
N_j	=	a set of remaining jobs, $N = N_i \cup N_j$
M_j	=	a set of machines that can process job j
$P_{j^*,k}$	=	the processing time of selected job j on machine k (hours)
$S_{i,j^*,k}$	=	the required setup time of machine k, when the machine is set up to process a selected job j and has a previous job i (hours).
D_{j^*}	=	the due date of selected job j (hours)
$C_{j,k}$	=	completion time of job j when processed on machine k (hours)

SCT (shortest job completion time): This rule simultaneously selects the next job $j \in N_j$ to be processed and machine $k \in M_j$ to process the job such that the completion time of job j is minimized, that is, SCT: $\min_{j\in N_j, k\in M_j}(S_{i,j,k} + P_{j,k})$. Note that, in most cases, the selected machine k processes Job i (the predecessor of Job j). This affects the set-up time $S_{i,j,k}$ of Job j on Machine k. That is, the selected job j may not be the job with the shortest processing time but rather the job with the shortest combined setup and processing times.

SC-LPT (shortest job completion time based on the longest processing time): This rule chooses the next job $j \in N_j$ that has the longest processing time first, for example, Job j^*. Then, it chooses machine $k \in M_{j^*}$ to process job j^* taking into account the setup time $S_{i,j^*,k}$, such that the job and machine selection results in the shortest completion time of job j. In other words, the rule is expressed as

$$\text{SC-LPT: } \min_{k\in M_{j^*}} (S_{i,j^*,k} + P_{j^*,k}) \Big| [P_{j^*,k} = \max_{j\in N_j, k\in M_j} P_{j,k}] .$$

SC-EDD (shortest job completion time based on the earliest due date): The rule takes the next Job $j \in N_j$ with the earliest due date first, for example, Job j^* After that, a Machine $k \in M_j$ is selected that results in the shortest completion time of Job j^*, given that the selected machine k is processing a preceding job i. That is,

$$\text{SC-EDD: } \min_{k\in M_{j^*}} (S_{i,j^*,k} + P_{j^*,k}) \Big| [D_{j^*} = \min_{j\in N_j, k\in M_j} D_j] .$$

In addition to the aforementioned rules, six combined rules are developed. Each consists of a pair of rules that are implemented sequentially with a specified rule-switching time t_s. For example, a combined rule [SC-EDD & SC-LPT: 200] first schedules jobs using the SC-EDD rule. When the completion time of the latest scheduled job exceeds $t_s = 200$ h, it switches to schedule jobs using the SC-LPT rule until all jobs are scheduled. The six combined rules include [SCT & SC-LPT: t_s], [SC-LPT & SCT: t_s], [SCT & SC-EDD: t_s], [SC-EDD & SCT: t_s], [SC-LPT & SC-EDD: t_s], and [SC-EDD & SC-LPT: t_s]. The steps of each rule are expressed as follows:

According to the illustration in Fig. 1, the algorithm sets $N_i = \{0\}$, $N_j = N$, and the parameters $P_{j,k}$, $S_{i,j,k}$, and D_j first. After that, it sets completion time of the dummy job $C_0 = 0$, and computes the completion time $C_{j,k} = P_{j,k}, \forall j \in N_j, \forall k \in M_j$ for all jobs on all possible machines. Then, the algorithm starts using Rule 1. After Job j^* is scheduled on machine k^*, the following are updated: the lists of scheduled jobs N_i and remaining jobs N_j. In addition, set $i = j^*$, and update the completion times of all remaining jobs that can be processed on machine k^* i.e. $C_{j,k^*} = S_{i,j,k^*} + P_{j,k^*}, \forall j \in N_j$, because the next job to be processed on this machine will follow Job $i = j^*$. After that, check whether all jobs are scheduled, that is, N_j is empty. If this is the case, the algorithm stops. Otherwise, check whether the maximum completion time of the scheduled job $C_{j,k}, \forall j \in N_i, \forall k \in M_j$ exceeds the rule-switching time t_s. If it does not, Rule 1 is continued. Otherwise, switch to Rule 2 and proceed until all the remaining jobs are scheduled. Note that when only one rule exists, t_s is set to a large value, which leads to no rule switching. The algorithm is implemented using Python programming language.

Numerical Example

A numerical example is provided to demonstrate the dispatching rules developed in the previous section. Suppose there are three jobs to be processed by two unrelated parallel machines with a sequence-dependent setup time. The processing times, setup times, and due dates of each job are listed in Table 2. Because jobs 1 and 2 are in the same job family, the setup times when one follows the other are relatively short. Job 3, on the other hand, is in a different job family from jobs 2 and 3; therefore, the setup times when either job 1 or job 2 succeeds or precedes job 3 are longer.

The steps of a single dispatching rule, SCT, are described below as an example, whereas the steps of using SC-LPT and SC-EDD are demonstrated in Appendix A.

Initialize set $N_i = \{0\}$, $N_j = N$, parameters $P_{j,k}$, $S_{i,j,k}$, and D_j

Set $i = 0$, and $C_i = 0$ (dummy job)

Compute $C_{j,k} = P_{j,k}, \forall j \in N_j, \forall k \in M_j$ (all jobs)

while $N_j \neq \{\}$ **do**

if $\max_{j \in N_i, k \in M_j} C_{j,k} \leq t_s$, Rule 1 is used and job and machine selection steps (1) to (3) are performed:

(1) Select job $j^* \in N_j$, and machine $k^* \in M_j$ to process job j^*.

(2) Add job j^* to N_i and remove job j^* from N_j.

(3) Set $i = j^*$ and update $C_{j,k^*} = S_{i,j,k^*} + P_{j,k^*}, \forall j \in N_j$.

else, switch to Rule 2 and perform job and machine selection steps (1) to (3).

end

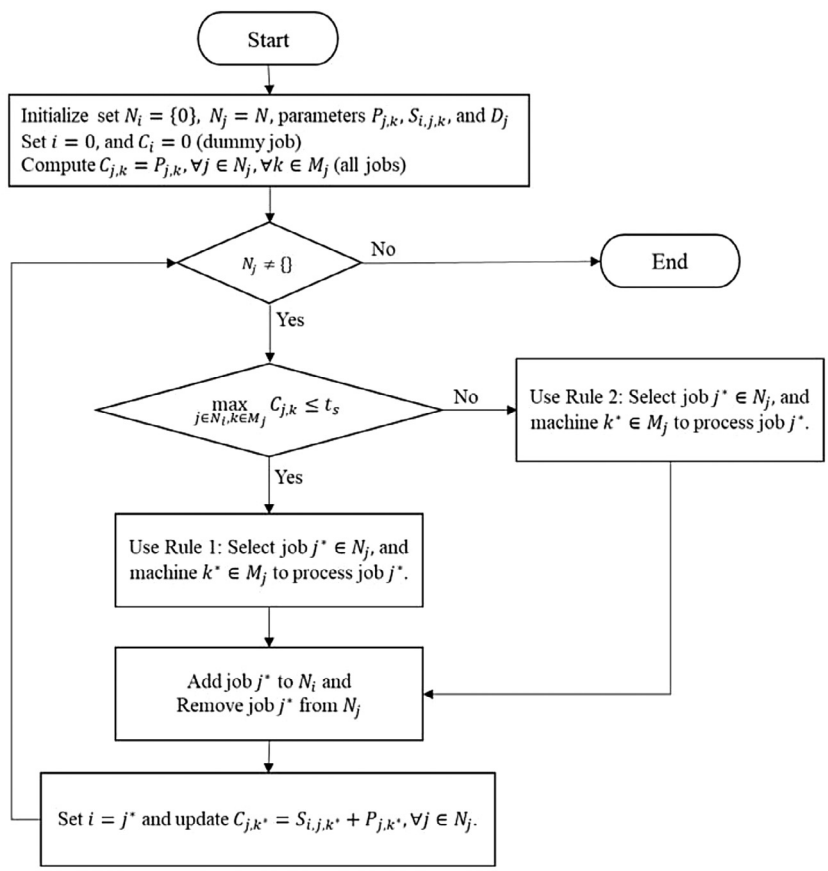


Fig. 1. Flow chart of the dynamic dispatching rule algorithm.

Table 2
Numerical example data.

j	$P_{j,1}$	$P_{j,2}$	D_j	$S_{i,j,1}$		j			$S_{i,j,2}$		j		
						1	2	3			1	2	3
1	9	6	30		1	0	0.5	5		1	0	0.5	3
2	22	16	20	i	2	0.5	0	5	i	2	0.5	0	3
3	28	22	32		3	5	5	0		3	3	3	0

SCT: This rule considers the job completion time as the main criterion. The rule proceeds as presented in Table 3.

From Table 3, in Step 1, Job 1 provides the least completion time for machine 2. Therefore, it is selected with a completion time of 6 h. In Step 2, the setup and completion times of jobs 2 and 3 on machine 2 are updated to reflect the previously scheduled job 1 on machine 2. This leads to the decision to schedule job 2 on machine 1 as the best choice, with the least completion time of 22 h. After updating the setup time and completion time of job 3 on machine 1 in step 3, job 3 is selected to be processed by machine 2, which gives a completion time of 31 h.

Table 3
Steps of SCT.

Step	j	D_j	$P_{j,1}$	$P_{j,2}$	$S_{i,j,1}$	$S_{i,j,2}$	$C_{j,1}$	$C_{j,2}$	j^*	M_{j^*}
1	**1**	30	9	6	0	0	9	**6**	1	2
	2	20	22	16	0	0	22	16		
	3	32	28	22	0	0	28	22		
2	**2**	20	22	16	0	0.5	**22**	22.5	2	1
	3	32	28	22	0	3	28	31		
3	**3**	32	28	22	5	3	55	**31**	3	2

Table 4
Steps of SCT & SC-LPT: $t_s = 5$.

Step	j	D_j	$P_{j,1}$	$P_{j,2}$	$S_{i,j,1}$	$S_{i,j,2}$	$C_{j,1}$	$C_{j,2}$	j^*	M_{j^*}
1	**1**	30	9	6	0	0	9	**6**	1	2
	2	20	22	16	0	0	22	16		
	3	32	28	22	0	0	28	22		
2	2	20	22	16	0	0.5	22	22.5	3	1
	3	32	**28**	22	0	3	**28**	31		
3	2	20	22	16	5	0.5	55	22.5	2	2

Table 5
Summary of results from four rules.

Rule	C_{max}	T	L
SCT	31	2	1
LPT	31	3	2
EDD	41	9	1
[SCT & SC-LPT]	28	2.5	1

This schedule resulted in a makespan of 31 h, a total tardiness of 2 h, and only job 2 is tardy.

Next, one of the combined rules, [SCT & SC-LPT: $t_s = 5$], which starts with the SCT rule and switches to the SC-LPT rule after 5 h, is described as an example. The rule proceeds as presented in Table 4.

First, using the SCT rule in Step 1, job 1 is selected to be processed on machine 2 with a completion time of 6 h. Subsequently, in Step 2, the heuristic switches to SC-LPT, which chooses job 3 to be processed on machine 1 with a completion time of 28 h. Finally, in Step 3, job 2 is processed on machine 2, which provides a completion time of 22.5 h. Using the combined rule, the makespan is 28 h, the total tardiness is 2.5 h, and only job 2 is tardy.

The performance of the four rules in the numerical example is summarized in Table 5. In general, it is observed that among the first three dispatching rules, SCT performed the best in this example. In addition, the solution from the combined rules can improve the makespan over the SCT rule, with a tradeoff of a 0.5 h increase in the total tardiness. This indicates the potential benefit of combining the dispatching rules.

5. Computational study

5.1. Small problem instances

The models described in Section 3 are implemented in ILOG CPLEX 20.1. The purpose of solving the small problem instances is to demonstrate the problem and the formulated models, as well as to investigate the characteristics of the problems with respect to the three system measures of performance. Based on our initial computational test, small problem instances with a size of up to three machines and ten jobs can provide insight into the problem and can be solved within a reasonable computational time. The data for the small problem instances are samples of real data. The source of the data is an industrial user, the largest steel pipe manufacturer in Thailand. Problem instances 1 and 2 are sample data from the production months with relatively low demand, whereas problem instances 3 and 4 are from the production months with high demand. Furthermore, instances 1 and 3 contain jobs that can be processed on some machines, whereas instances 2 and 4 include jobs that can be processed on any machine. Problem instance 1 is presented below, and the other instances are provided in the Appendix.

$$P_{j,k} = \begin{bmatrix} 66.90 & 11.80 & 26.80 & 6.45 & 30.74 & 11.23 & 4.51 & 5.09 & 9.70 & 3.18 \\ 100 & 100 & 33.49 & 8.07 & 38.43 & 14.03 & 100 & 100 & 12.13 & 3.97 \\ 66.90 & 11.80 & 26.80 & 6.45 & 30.74 & 11.23 & 4.51 & 5.09 & 100 & 100 \end{bmatrix}$$

$$D_j = [79\ 30\ 60\ 25\ 57\ 38\ 11\ 7\ 42\ \ 4]$$

$$NP_{j,k} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$S_{i,j,1} = \begin{bmatrix} 0 & 3 & 5 & 5 & 5 & 5 & 4 & 4 & 5 & 5 \\ 3 & 0 & 5 & 5 & 5 & 5 & 4 & 4 & 5 & 5 \\ 5 & 5 & 0 & 0.5 & 0.5 & 3 & 5 & 5 & 5 & 5 \\ 5 & 5 & 0.5 & 0 & 0.5 & 3 & 5 & 5 & 5 & 5 \\ 5 & 5 & 0.5 & 0.5 & 0 & 3 & 5 & 5 & 5 & 5 \\ 5 & 5 & 3 & 3 & 3 & 0 & 5 & 5 & 5 & 5 \\ 4 & 4 & 5 & 5 & 5 & 5 & 0 & 0.5 & 5 & 5 \\ 4 & 4 & 5 & 5 & 5 & 5 & 0.5 & 0 & 5 & 5 \\ 5 & 5 & 5 & 5 & 5 & 5 & 5 & 5 & 0 & 0.5 \\ 5 & 5 & 5 & 5 & 5 & 5 & 5 & 5 & 0.5 & 0 \end{bmatrix}$$

$$S_{i,j,2} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0.5 & 3 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0.5 & 0 & 0.5 & 3 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0.5 & 0.5 & 0 & 3 & 0 & 0 & 5 & 5 \\ 0 & 0 & 3 & 3 & 3 & 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 5 & 5 & 5 & 0 & 0 & 0 & 0.5 \\ 0 & 0 & 5 & 5 & 5 & 5 & 0 & 0 & 0.5 & 0 \end{bmatrix}$$

$$S_{i,j,3} = \begin{bmatrix} 0 & 3 & 6 & 6 & 6 & 6 & 4 & 4 & 0 & 0 \\ 3 & 0 & 6 & 6 & 6 & 6 & 4 & 4 & 0 & 0 \\ 6 & 6 & 0 & 4 & 4 & 3 & 6 & 6 & 0 & 0 \\ 6 & 6 & 4 & 0 & 0.5 & 4 & 6 & 6 & 0 & 0 \\ 6 & 6 & 4 & 0.5 & 0 & 4 & 6 & 6 & 0 & 0 \\ 6 & 6 & 3 & 4 & 4 & 0 & 6 & 6 & 0 & 0 \\ 4 & 4 & 6 & 6 & 6 & 6 & 0 & 0.5 & 0 & 0 \\ 4 & 4 & 6 & 6 & 6 & 6 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The number of decision variables, binary variables, and computational times for each model are listed in Table 6.

5.1.1. Optimal solutions

The optimal solution from M1 for problem instance P1 is presented in Table 7 and Fig. 2. Appendix B presents the other optimal solutions for M2 and M3.

Fig. 2 illustrates the three machines working on ten jobs. Each machine sequentially processes a job. Machine 1 processes five jobs, starting with jobs 3, 6, 2, 7, and 8. Machine 2 serves four jobs, beginning with jobs 10, 9, 5, and 4. The last machine, Machine 3, operated only one job, 1. Fig. 2 also demonstrates that the total time for the setup is 18.5 h. The optimal makespan is 71.93 h. Furthermore, these two machines finish jobs 8, 7, 2, 6, 4, and 5 after the due date. The total tardiness is 197.75 h.

5.1.2. Pareto solution

M4 is used to find the comprise solutions by minimizing the makespan while fixing the value of the total tardiness (T) and the number of tardy jobs (L). It is necessary to create a payoff table from

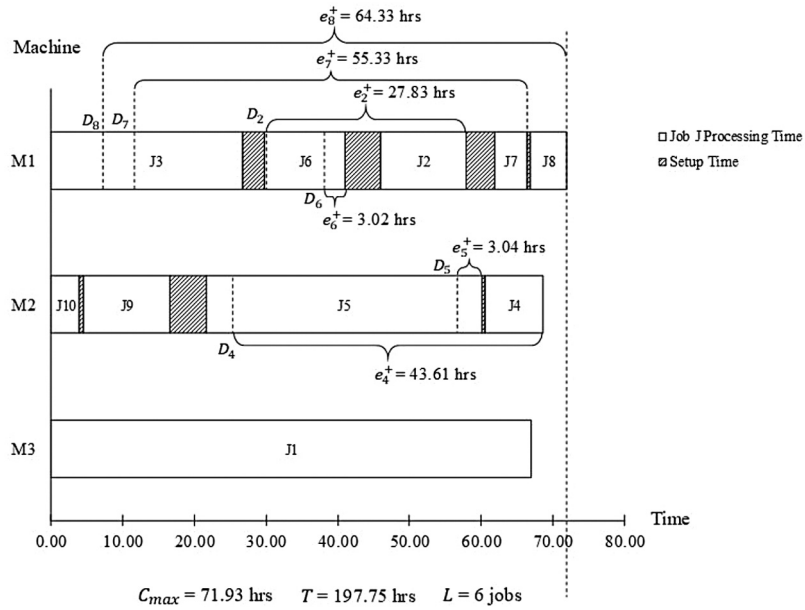


Fig. 2. Job sequence at each machine from minimizing the makespan.

Table 6
The decision variables, binary variables, and computational times.

Model	Decision variable	Binary variable	Computational time			
			P1	P2	P3	P4
M1 — Minimize the makespan	342	330	22 s	4 min	15 s	8 min
M2 — Minimize the total tardiness	362	330	8 s	29 s	9 s	36 s
M3 — Minimize the number of tardy jobs	372	340	7 s	9 s	8 s	10 s
M4 — Determine compromise solutions	372	340	21 min	4 h 39 min	15 min	5 h 40 min

Table 7
Completion time, due date, and tardiness of each job from minimizing the makespan.

Job	Completion time	Due date	Tardiness
1	66.90	79	0
2	57.83	30	27.83
3	26.80	60	0
4	68.61	25	43.61
5	60.04	57	3.04
6	41.02	38	3.02
7	66.33	11	55.33
8	71.93	7	64.93
9	16.61	42	0
10	3.98	4	0

Table 8
Payoff table for P1.

	C_{max}	T	L
Min C_{max}	71.93	**197.75**	**6**
Min T	81.00	**11.23**	4
Min L	127.95	48.95	**1**

the optimal solutions for M1, M2, and M3. The payoff table for P1 is presented in Table 8.

This payoff table shows that the smallest tardiness is 11.23 h from M2 and the largest tardiness is 197.75 h from M1. The largest number of tardy jobs is six from M1, and the smallest number of tardy jobs is one from M3. From the payoff table, the range of L is [1, 2, 3, 4, 5, 6]. Next, the range of tardiness is set to 20 grid points with a step size of (197.75 − 11.23)/20 = 9.33. Hence, the range of tardiness, T = [11.23, 20.56, . . . , 197.75]. After solving M4 for 126 runs, there are nine solutions, the so-called Pareto solutions, that are not dominated by other solutions in all three measures of performance. Table 9 shows the set of Pareto optimal solutions for the four problems illustrated in Fig. 3.

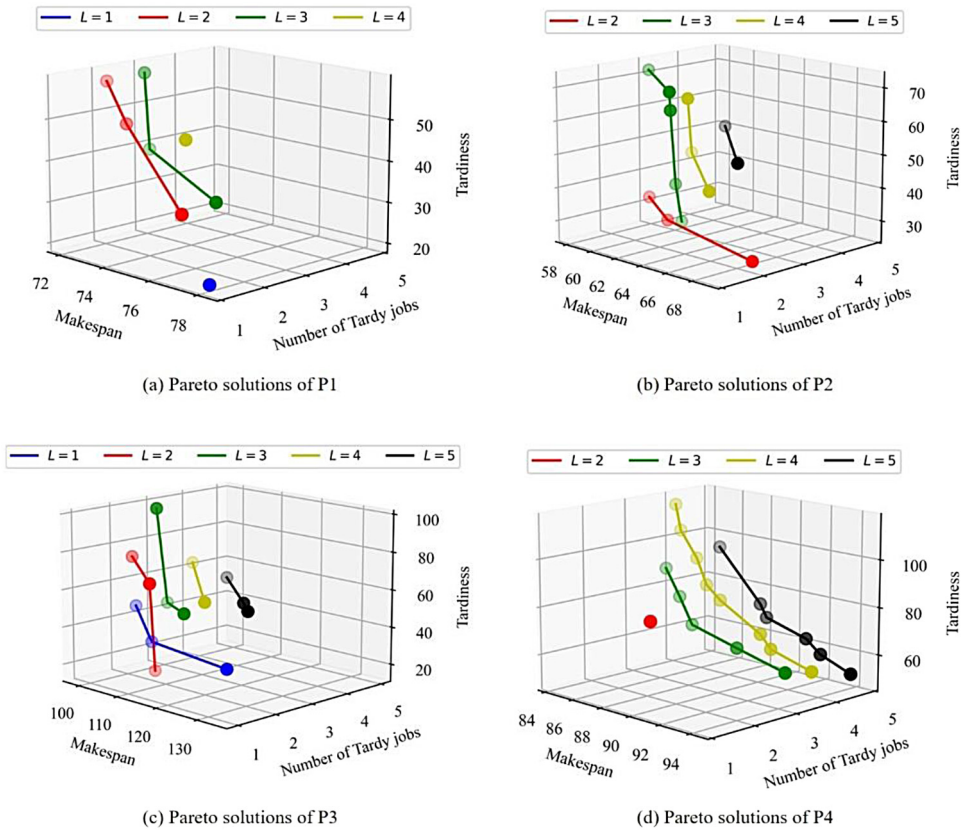
From Fig. 3, it can be observed that given a number of tardy jobs, the total tardiness decreases as the makespan increases. Similarly, given the total tardiness, the makespan decreased as the number of tardy jobs increased. The same observation can be made for the relationship between the number of tardy jobs and the total tardiness, given a makespan.

To evaluate the quality of the Pareto solutions, hypervolume percentages for the four problem instances are computed as shown in Table 10. The Pareto solutions from AUGMECON method range from 74% (P3) to 91% (P1).

Among the Pareto solutions, a decision maker can choose one solution by specifying the weight of importance for each measure of performance and applying the min–max normalization as follows:

$$\tilde{x}_i = \frac{x_i - \min_i \{x_i\}}{\max_i \{x_i\} - \min_i \{x_i\}}$$

Suppose that the decision maker assigns the following weights to the three measures of performance: $W_{C_{max}} = 0.5, W_T = 0.4, W_L = 0.1$. For P1, the minimum C_{max} is 71.93 and the maximum C_{max} is 78.35, the minimum T is 20.56, the maximum T is 57.86, the minimum L is 1, and the maximum L is 4. All Pareto solution measures of performance are normalized and weighted to obtain the total normalized score, as shown in Table 11. The solution with the lowest total normalized score is selected by the decision maker. For P1, Solution 6 with $C_{max} = 72.14$,



(a) Pareto solutions of P1

(b) Pareto solutions of P2

(c) Pareto solutions of P3

(d) Pareto solutions of P4

Fig. 3. Pareto solutions of small problem instances.

Table 9
Pareto solutions.

P1			P2			P3			P4		
C_{max}	T	L	C_{max}	T	L	C_{max}	T	L	C_{max}	T	L
78.35	20.56	1	69.14	26.90	2	134.36	37.49	1	88.24	76.65	2
75.41	29.88	2	62.66	32.47	2	116.18	41.68	1	94.42	60.48	3
73.01	48.53	2	61.20	38.03	2	112.28	58.43	1	91.27	65.87	3
72.14	57.86	2	**60.56**	**26.90**	**3**	**107.26**	**16.55**	**2**	**88.24**	**71.26**	**3**
75.08	29.88	3	60.07	38.03	3	105.98	62.61	2	87.38	82.04	3
72.14	**39.21**	**3**	59.67	60.31	3	101.46	75.18	2	86.43	92.82	3
71.93	57.86	3	59.60	65.87	3	105.31	41.68	3	93.56	55.09	4
71.93	39.21	4	57.90	71.44	3	100.93	45.86	3	90.86	60.48	4
			59.67	32.47	4	98.25	96.11	3	90.16	65.87	4
			58.21	43.60	4	101.46	41.68	4	87.38	76.65	4
			57.90	60.31	4	98.25	62.61	4	86.43	82.04	4
			58.95	38.03	5	103.88	33.30	5	85.74	92.82	4
			57.90	49.17	5	102.75	37.49	5	84.59	103.6	4
						98.25	50.05	5	84.24	114.38	4
									93.61	49.7	5
									91.61	55.09	5
									90.64	60.48	5
									87.93	65.87	5
									87.49	71.26	5
									84.59	92.82	5

Table 10
Hypervolume of Pareto solutions.

	P1			P2			P3			P4		
	C_{max}	T	L	C_{max}	T	L	C_{max}	T	L	C_{max}	T	L
Origin point	71.93	11.23	1	57.90	21.33	2	98.25	16.55	1	84.42	49.70	2
Reference point	127.95	197.75	6	84.63	132.96	6	134.36	93.70	5	133.50	157.50	5
Hypervolume indicator	48,057.58			10,768.40			8500.54			12,971.95		
Bounding hypervolume	52,244.25			11,935.48			11,491.65			15,880.23		
Hypervolume percentage	91.99%			90.22%			73.97%			81.69%		

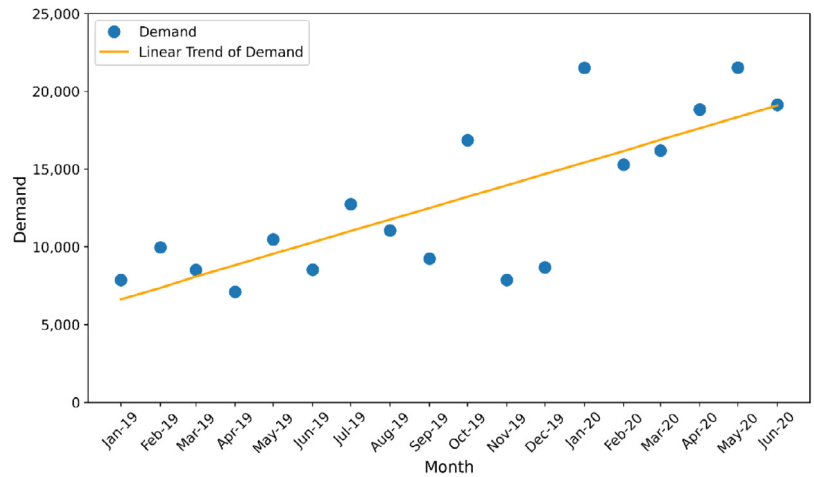


Fig. 4. 18-month demand.

Table 11
The total normalized score of P1's Pareto solutions.

Solution number	C_{max}	T	L	Total normalized score
1	0.5000	0.0000	0.0000	0.5000
2	0.2706	0.1000	0.0333	0.4040
3	0.0838	0.3000	0.0333	0.4172
4	0.0164	0.4000	0.0333	0.4498
5	0.2455	0.1000	0.0667	0.4122
6	0.0164	0.2000	0.0667	**0.2831**
7	0.0000	0.4000	0.0667	0.4667
8	0.0000	0.2000	0.1000	0.3000

$T = 39.21$, and $L = 3$ is selected, as shown in Table 9. The selected solutions obtained by applying the same procedure to P2, P3, and P4 are presented in Table 12.

5.1.3. Performance of the heuristic methods on the small problem instances

The performance of the heuristics is evaluated using small problem instances with known Pareto solutions. Each best-found solution from the heuristics is compared with one of the Pareto solutions closest to the best-found heuristic solution. Table 13 presents a comparison of the results for P1. For example, the first heuristic solution obtained from [SC-EDD & SC-LPT: 60] had a $L = 2$, $C_{max} = 106.59$, and $T = 29.19$. Note that the closest Pareto solution is chosen among the solutions with the same L, which results in a solution with $C_{max} = 75.41$, and $T = 29.88$. The %off Pareto solutions for C_{max} and T are 41.35% and −2.31%, respectively. This indicates that the heuristic solution is worse than the selected Pareto solution in terms of C_{max}, but outperforms the Pareto solution in terms of T, which implies that the heuristic solution is not inferior to the Pareto solution. Similar arguments are applied for the second heuristic solution from [SC-EDD & SC-LPT: 30] with $L = 3$, $C_{max} = 81$, and $T = 18.02$. However, the best-found heuristic solution, where $L = 4$, $C_{max} = 77.88$, and $T = 41.78$, is inferior to the closest Pareto solution. The results of the heuristics for P2, P3, and P4 can be found in Appendix C.

5.2. Large problem instances

Large problem instances are created using 18 months of real production data from January 2019 to June 2020. Each problem instance is a month of production data. In total, 413 products are classified into 26 product families. These products are steel pipes with (1) different thicknesses, which are categorized as being in the same family, that is, requiring a short setup time; (2) different diameters, medium setup times depending on the change in diameters; or (3) different shapes: round, rectangular, or square, which are classified as being in different families, that is, requiring a long setup time. The monthly demand data contain customer orders ranging from 1359 to 3639 orders. Each order contains up to 40 products. Customer orders requesting the same products are aggregated into the same job, that is, production orders. On average, the number of jobs in each month ranged from 244 to 298.

The production line has ten unrelated parallel machines with varying capabilities in terms of the product families that each machine can process and the production rate. The setup times for products in the same family but with different thicknesses ranged from 20 to 40 min. However, products in the same family with different diameters or products in different families require setup times ranging from 3 to 11 h, depending on the sequence of jobs, that is, changing diameters requires a shorter setup time than changing shape. Statistical summary of the large instances is presented in Table 14.

Several months of data are tested to demonstrate the effectiveness of the proposed heuristics in various demand scenarios. Naturally, there is some variation in monthly demand and an increasing trend in demand over time due to business growth (see Fig. 4). Specifically, the monthly demand in 2019 is lower than that in 2020, except in October 2019. Interestingly, there is growing demand despite the Covid-19 pandemic. The problem instances are grouped using the k-mean clustering method to characterize the demand scenarios. Based on the analysis, the 18 months of demand data can be divided into two clusters: low demand and high demand. The number of clusters is chosen based on the highest silhouette score [44], as shown in Table 15.

Each problem instance is tested with three single dispatching rules and six combined rules in combination with six rule-switching times. In a month, there are up to 25 working days, eight hours per shift, two shifts per day, and up to five hours of overtime, which results in a maximum of 525 production hours. Therefore, the rule-switching times are set between 200 and 450 h with a step size of 50 h, which is 200, 250, 300, 350, 400, and 450. In summary, the computational experiment tests $(3 + 6 \times 6) = 39$ rules on 18 problem instances and three measures of performance, including makespan, tardiness, and number of tardy jobs.

5.2.1. Single dispatching rule results

First, the average performance of the three single dispatching rules is reported for both the demand scenarios in Table 16. Statistical analysis, including analysis of variance (ANOVA) and Tukey's test, is performed on each measure of performance treated as a response variable, the three rules as experimental factors, and demand scenarios as blocks. According to the ANOVA, the performances of the three

Table 12
The selected one solution of the set of Pareto solutions for the four problems.

P1			P2			P3			P4		
C_{max}	T	L	C_{max}	T	L	C_{max}	T	L	C_{max}	T	L
72.14	39.208	3	60.56	26.90	3	107.26	16.55	2	88.24	71.26	3

Table 13
The solutions from heuristic methods and pareto, and %-Off Pareto solutions for P1.

Rules	Heuristics solutions			Pareto solutions			%-Off Pareto solutions	
	L	C_{max}	T	L	C_{max}	T	C_{max}	T
[SC-EDD & SC-LPT: 60]	2	106.59	29.19	2	75.41	29.88	41.35%	−2.31%
[SC-EDD & SC-LPT: 30]	3	81	18.02	3	75.08	29.88	7.88%	−39.69%
[SC-LPT & SCT: 10]	4	77.88	41.78	4	71.93	39.21	8.27%	6.55%

Table 14
Statistical summary of large problem instances.

Dataset	Demand scenario	No. of jobs	Quantity				No. of jobs that can be processed by k machines						
			Avg	SD	Max	Min	k = 1	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7
1	Low	839	31.75	44.75	418.13	0.01	78	4	52	19	50	37	7
2	Low	809	40.36	50.09	369.93	0.02	87	4	43	16	61	36	–
3	Low	838	33.65	53.68	538.08	0.06	87	3	46	16	63	38	–
4	Low	815	29.51	41.30	323.79	0.03	80	2	43	16	62	38	–
5	Low	868	41.09	71.01	711.91	0.13	82	3	47	14	66	43	–
6	Low	886	33.08	51.64	374.96	0.03	82	3	48	16	64	45	–
7	Low	914	47.76	68.32	606.10	0.05	85	4	51	16	62	49	–
8	Low	938	40.93	59.10	587.33	0.06	84	3	51	16	65	51	–
9	Low	917	34.76	55.66	591.82	0.03	81	4	53	17	65	46	–
10	High	952	60.84	89.21	828.87	0.06	87	4	53	16	68	49	–
11	Low	922	44.26	61.28	480.95	0.06	88	4	53	14	67	46	–
12	Low	941	31.59	43.69	336.00	0.07	85	4	57	14	69	46	–
13	High	976	74.67	110.13	1018.96	0.05	92	4	59	16	67	50	–
14	High	932	54.99	86.00	756.49	0.33	91	4	56	15	67	45	–
15	High	998	55.27	94.90	947.90	0.08	93	3	59	15	76	47	–
16	High	1025	63.22	102.91	981.80	0.03	91	4	60	17	78	48	–
17	High	958	76.85	128.43	1433.85	0.10	90	3	52	16	72	47	–
18	High	1029	64.86	99.60	893.22	0.04	90	3	55	18	78	51	–

Table 15
Silhouettes score.

The number of clusters	2	3	4	5	6	7	8	9	10
Silhouettes score	0.76	0.72	0.69	0.75	0.71	0.73	0.68	0.6	0.58

Table 16
Average performance measures of dispatching rules.

Demand scenario	Rule	C_{max}	T	L
Low	SCT	466.99^{C}	$3{,}915.75^{B}$	43.36^{B}
	SC-EDD	629.86^{B}	$3{,}657.31^{B}$	51.18^{B}
	SC-LPT	907.15^{A}	$26{,}312.50^{A}$	113.64^{A}
High	SCT	637.97^{C}	$8{,}770.64^{B}$	67.14^{B}
	SC-EDD	827.20^{B}	$11{,}336.01^{B}$	70.00^{B}
	SC-LPT	1112.51^{A}	$55{,}823.23^{A}$	179.00^{A}

Note: Average measures that do not share a letter are significantly different.

rules are statistically different. In addition, Tukey's comparison results indicate that the performance of SCT and SC-EDD rules is statistically better than that of the SC-LPT rule in all three measures and in both demand scenarios. Therefore, SC-LPT is eliminated from further analysis. For SCT and SC-EDD, it is observed that (1) SCT provides statistically better results in terms of the makespan (as expected), (2) SCT provides relatively better results in terms of the number of tardy jobs, and (3) SC-EDD yields better results in terms of total tardiness under the low-demand scenario but worse results in the high-demand scenario. Note that the performances of the two rules are not statistically different in terms of the total tardiness and number of tardy jobs.

5.2.2. Combined dispatching rule results

For the six combined rules, there could be effects of rule-switching time and demand scenarios as well as their interactions. Therefore, the six combined rules are treated as a categorical factor with six levels, and the rule-switching time is treated as a covariate. The 18 problem instances, split into two demand scenarios, are treated as another factor with two levels. Regression models for each performance measure are fitted to determine the effective rule(s) and appropriate rule-switching time for each demand scenario. The results of the statistical analysis used to evaluate the factor effects are shown in Tables 17–19.

From Table 17, because the three-factor interaction is significant, the effect of the dispatching rules on the makespan clearly depends on the rule-switching time and demand scenarios. The relationship between the makespan and rule-switching time for each rule under a demand scenario is shown in Table 18 and plotted in Fig. 5. From Fig. 5, the effective rules for reducing makespan are [SCT & SC-LPT: 450] for the high-demand scenario, and [SCT & SC-EDD: 450], [SCT & SC-LPT: 450] for the low-demand scenario. Note that these rules use SCT in a much larger proportion of the production time in a month before switching to other rules.

A similar analysis is performed on total tardiness and number of tardy jobs. According to the ANOVA results, the three-factor interaction of dispatching rule, rule-switching time, and demand scenario is also significant for both responses. Based on the analysis of the three measures of performance, the effective rules for all under the two demand scenarios are summarized in Table 19. Among the 36 possible rules (six combined rules and six rule-switching times), there are three effective combined rules: (1) [SCT & SC-LPT: 450] for makespan and the number of tardy jobs in both demand scenarios; (2) [SCT & SC-EDD: 450] for makespan in low demand, total tardiness in high demand, and the

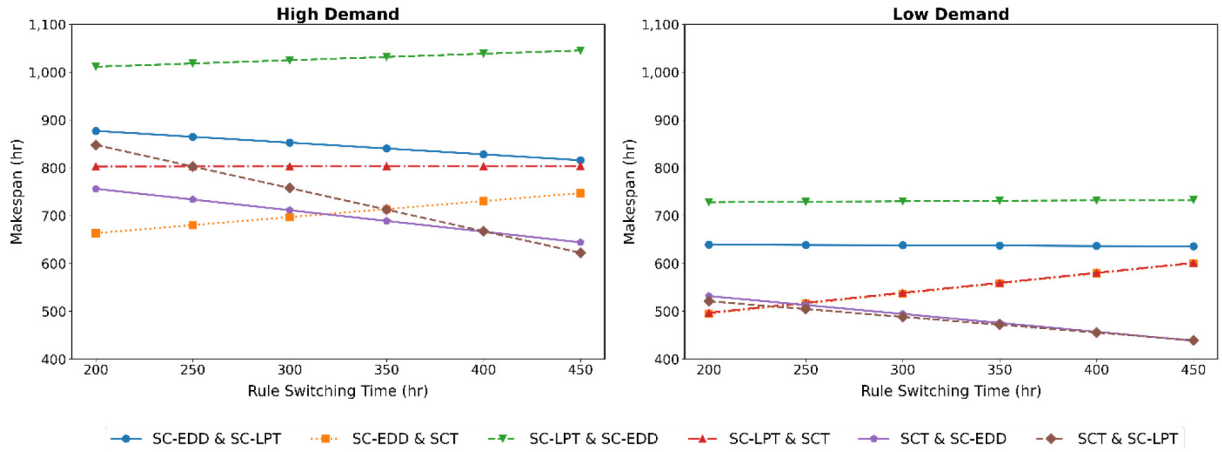


Fig. 5. Regression models of different dispatching rules on the makespan.

Table 17
ANOVA for makespan.

Source	DF	Seq SS	Seq MS	F-Value	P-Value
Regression	23	15,219,517	661,718	169.23	0.000
Dispatching rule	5	6,023,192	1,204,638	308.08	0.000
Rule-switching time	1	15,919	15,919	4.07	0.044
Demand scenario	1	8,238,496	8,238,496	2106.94	0.000
Dispatching rule × Rule-switching time	5	532,531	106,506	27.24	0.000
Dispatching rule × Demand scenario	5	300,433	60,087	15.37	0.000
Rule-switching time × Demand scenario	1	49,940	49,940	12.77	0.000
Dispatching rule × Rule-switching time × Demand scenario	5	59,006	11,801	3.02	0.011
Error	624	2,439,949	3,910		
Lack-of-fit	48	49,133	1,024	0.25	1.000
Pure error	576	2,390,816	4,151		
Total	647	17,659,467			

Table 18
Regression equations for makespan.

Demand scenario	Rule	Regression equation
High	SC-EDD & SC-LPT	Makespan = 926.1 − 0.245Time
High	SC-EDD & SCT	Makespan = 596.4 + 0.335Time
High	SC-LPT & SC-EDD	Makespan = 984.1 + 0.136Time
High	SC-LPT & SCT	Makespan = 801.8 + 0.004Time
High	SCT & SC-EDD	Makespan = 845.7 − 0.448Time
High	SCT & SC-LPT	Makespan = 1028.2 - 0.902Time
Low	SC-EDD & SC-LPT	Makespan = 642.2 − 0.014Time
Low	SC-EDD & SCT	Makespan = 410.9 + 0.4207Time
Low	SC-LPT & SC-EDD	Makespan = 723.8 + 0.0205Time
Low	SC-LPT & SCT	Makespan = 413.5 + 0.4173Time
Low	SCT & SC-EDD	Makespan = 606.6 − 0.3738Time
Low	SCT & SC-LPT	Makespan = 586.4 − 0.327Time

Table 19
Effective combined dispatching rules.

Measures	Demand scenario	
	High	Low
Makespan	[SCT & SC-LPT: 450]	[SCT & SC-EDD: 450] [SCT & SC-LPT: 450]
Total tardiness	[SCT & SC-EDD: 450]	[SC-EDD & SCT: 200]
Number of tardy jobs	[SCT & SC-EDD: 450] [SCT & SC-LPT: 450] [SC-EDD & SCT: 200]	[SCT & SC-EDD: 450] [SCT & SC-LPT: 450] [SC-EDD & SCT: 200]

number of tardy jobs in both demand scenarios; and (3) [SC-EDD & SCT: 200] for the total tardiness in low demand and the number of tardy jobs in both demand scenarios.

5.3. Selection of dispatching rule under multiple measures of performance

The effective dispatching rules (two single and three combined rules) and their performances on the three measures are summarized in Table 20. The selection of dispatching rules becomes a multi-criteria decision-making problem. A closer look reveals that some rules, although effective, are inferior to others. Specifically, under high demand, the [SCT & SC-LPT: 450] is outperformed by SCT in all three measures. Therefore, [SCT & SC-LPT: 450] is eliminated for further analysis under high demand. As a result, four rules remain for the high-demand scenario. Similarly, under low demand, SCT and [SCT & SC-LPT: 450] are eliminated by [SCT & SC-EDD: 450] and SC-EDD by [SC-EDD & SCT: 200]. In other words, there are only two rules, [SC-EDD & SCT: 200] and [SCT & SC-EDD: 450], to be considered under low demand.

For the multi-criteria decision-making problem, the weight of importance assigned to each measure is critical to the rule selection process. Although these weights are usually specified by a decision maker (DM), we chose to perform rule selection analysis under exhaustive enumerations of the three weights. The purpose is to obtain the conditions specified by the relative weights assigned to the measures, where each dispatching rule is the most effective. The weights for the makespan, total tardiness, and number of tardy jobs are denoted by w_C, w_T, and w_L, respectively. Their values are varied from 0.01 to 0.98, under the condition that the weights add up 1, i.e., w_C, w_T, and $w_L \in [0.01, 0.98]$ and $w_C + w_T + w_L = 1$. Because the three measures are numerical, an appropriate selection method is the technique for order preference by similarity to ideal solution (TOPSIS). The steps of TOPSIS implemented in this study are described below [45].

Step 1: Construct a matrix of performance measure x_{ab} for dispatching rule a and measure of performance b. For our problem, all three

Table 20
Effective dispatching rules.

Demand scenario	Dispatching rule	C_{max}	T	L	Eliminated by
High	SCT	637.97	8770.64	67.14	
High	SC-EDD	827.20	6223.62	70	
High	[SC-EDD & SCT: 200]	660.77	8003.79	66.43	
High	[SCT & SC-LPT: 450]	644.43	11,093.70	74.43	SCT
High	[SCT & SC-EDD: 450]	652.47	7565.10	68.43	
Low	SCT	466.99	3915.75	43.36	[SCT & SC-EDD: 450]
Low	SC-EDD	629.86	3657.31	51.18	[SC-EDD & SCT: 200]
Low	[SC-EDD & SCT: 200]	491.22	2429.83	41.09	
Low	[SCT & SC-LPT: 450]	453.22	3908.48	42.73	[SCT & SC-EDD: 450]
Low	[SCT & SC-EDD: 450]	452.94	3795.79	42.63	

Table 21
Numerical example for TOPSIS.

Step 1: x_{ab}				Steps 2–3: r_{ab}, v_b^+, v_b^-				Step 4: S_a^-, S_a^+			Step 5: C_a^*	
	b				b							
a	1	2	3	a	1	2	3	a	S_a^-	S_a^+	a	C_a^*
1	491.22	2429.83	41.09	1	0.92	1	1	1	0.19	0.05	1	0.81
2	452.94	3795.79	42.63	2	1	0.64	0.96	2	0.05	0.19	2	0.19
$\tilde{x}_b$	452.94	2429.83	41.09	v_b^+	1	1	1					
				v_b^-	0.92	0.64	0.96					

measures are considered as *cost* attributes; the lower the value of x_{ab}, the better is the performance.

Step 2: Compute $r_{ab} = \frac{\tilde{x}_b}{x_{ab}}$ as a normalized value of x_{ab}, $\tilde{x}_b = \min_a\{x_{ab}\}$. After normalization, the higher the value of r_{ab}, the better the performance.

Step 3: Find a positive ideal solution (v_b^+) and negative ideal (v_b^-) solutions for each measure of performance, where $v_b^+ = \max_a\{r_{ab}\}$, and $v_b^- = \min_a\{r_{ab}\}$.

Step 4: Compute *separate* measures for each dispatching rule a. Separate measures from the positive ideal solution are computed $S_a^+ = \sqrt{\sum_b w_b\left(r_{ab} - v_b^+\right)^2}$, and from the negative ideal solution $S_a^- = \sqrt{\sum_b w_b\left(r_{ab} - v_b^-\right)^2}$.

Step 5: Calculate the relative closeness to the ideal solution $C_a^* = \frac{S_a^-}{S_a^+ + S_a^-}$ for dispatching rule a. The greater the relative closeness, the better the performance.

Numerical example to compare the two rules, [SC-EDD & SCT: 200] and [SCT & SC-EDD: 450], for the low-demand scenario with $w_C = 0.36$, $w_T = 0.3$, and $w_L = 0.34$. Note that $a = 1$ denotes [SC-EDD & SCT: 200]; 2 denotes [SCT & SC-EDD: 450]; and $b = 1$, 2, and 3 denote the makespan, total tardiness, and number of tardy jobs, respectively. The calculation results are presented in Table 21. From the table, [SC-EDD & SCT: 200] is more effective.

After repeating TOPSIS for many combinations of weights, the conditions for each dispatching rule to be the most effective are as follows:

Low demand scenario: There are two rules to consider: [SC-EDD & SCT: 200], which gives the minimum total tardiness and minimum number of tardy jobs, and [SCT & SC-EDD: 450], which gives the minimum makespan. [SCT & SC-EDD: 450] is the most effective under the following conditions:

1. $w_C \geq 0.88$, $w_T = 0.04$, and $w_L \leq 0.08$
2. $w_C \geq 0.70$, $w_T = 0.03$, and $w_L \leq 0.27$
3. $w_C \geq 0.53$, $w_T = 0.02$, and $w_L \leq 0.45$
4. $w_C \geq 0.36$, $w_T = 0.01$, and $w_L \leq 0.63$

Otherwise, [SC-EDD & SCT: 200] is the most effective rule.

These conditions indicate that [SCT & SC-EDD: 450] is the most effective when the DM places little importance on total tardiness. In other words, [SC-EDD & SCT: 200] is dominant when $w_T \geq 0.05$. In addition, even though the DM gives little weight to the total tardiness, he or she must give enough weight to the makespan for [SCT & SC-EDD: 450] to be the more effective rule according to the four conditions.

High demand scenario: For the four considered rules, SCT, SC-EDD, [SCT & SC-EDD: 450], and [SC-EDD & SCT: 200]. The conditions are,

SCT is most effective under the conditions, $0.22 \leq w_C \leq 0.98$, $w_T = 0.01$, and $0.01 \leq w_L \leq 0.77$, where the makespan or number of tardy jobs is dominant in terms of their weights of importance.

[SC-EDD & SCT: 200] is the most effective under the following conditions: These conditions are applicable for various cases when the weight of the makespan is up to 29% and the weight of the total tardiness is relatively small (up to 7%).

1. $0.01 \leq w_C \leq 0.07, 0.01 \leq w_T \leq 0.07, 0.86 \leq w_L \leq 0.98$, and $w_T - w_C \leq 0.03$
2. $0.08 \leq w_C \leq 0.09, 0.01 \leq w_T \leq 0.06$, and $0.85 \leq w_L \leq 0.91$
3. $0.10 \leq w_C \leq 0.12, 0.01 \leq w_T \leq 0.05$, and $0.83 \leq w_L \leq 0.89$
4. $0.13 \leq w_C \leq 0.16, 0.01 \leq w_T \leq 0.04$, and $0.80 \leq w_L \leq 0.86$
5. $0.17 \leq w_C \leq 0.21, 0.01 \leq w_T \leq 0.03$, and $0.76 \leq w_L \leq 0.82$
6. $0.22 \leq w_C \leq 0.29, w_T = 0.02$, and $0.69 \leq w_L \leq 0.76$

[SCT & SC-EDD: 450] is the most effective under the four conditions listed below. These conditions indicate cases when the weight of the total tardiness is up to 50%.

1. $w_C \geq 0.49, w_T \geq 0.02$, and $w_L \leq 0.49$
2. $0.30 \leq w_C \leq 0.48, 0.02 \leq w_T \leq 0.50, 0.02 \leq w_L \leq 0.68$, and $(w_T - w_C) \leq 0.02$
3. $0.24 \leq w_C \leq 0.29, 0.03 \leq w_T \leq 0.31, 0.40 \leq w_L \leq 0.73$, and $(w_T - w_C) \leq 0.03$
4. $0.07 \leq w_C \leq 0.23, 0.03 \leq w_T \leq 0.24, 0.53 \leq w_L \leq 0.85$, and $(w_T - w_C) \leq 0.01$

Otherwise, the SC-EDD is effective when the weight of the total tardiness is greater than 50%.

5.4. Discussion

The findings of this research provide contributions to the production scheduling research area, particularly in the context of unrelated parallel machines with machine and job sequence-dependent setup times. The results are in line with previous research that emphasizes the complexity of scheduling problems in manufacturing environments. The classification of parallel machines into identical, uniform, and

unrelated categories has been well-documented in the literature [1–3]. This approach builds on this foundation by specifically addressing the challenges posed by machine and job sequence-dependent setups, which have been less frequently explored.

Performance of the modified dispatching rules, i.e., SPT, EDD, and LPT, demonstrates a clear benefit over traditional methods. Previous studies have shown that these rules can yield effective scheduling solutions under certain conditions [4,5]. Our research extends this understanding by incorporating machine and job sequence-dependent setup times, thereby enhancing the applicability of these rules in real-world scenarios. Moreover, the statistical analysis conducted, including ANOVA and Tukey's test, confirms that the performance of the SCT and SC-EDD rules is superior to that of the SC-LPT rule across various performance measures and demand scenarios. This finding is consistent with earlier work that suggests the importance of rule selection based on specific operational contexts [6]. The ability to adapt dispatching rules to the unique characteristics of each scheduling problem is crucial for improving production efficiency.

The multi-criteria decision-making analysis further enriches these findings by demonstrating how different performance measures can be prioritized based on the decision-maker's preference. This aspect of the research resonates with the literature that supports a tailored approach to scheduling, where the relative importance of makespan, tardiness, and job completion times can vary significantly depending on the production environment [9]. This study highlights that combined dispatching rules, such as [SCT & SC-LPT] and [SCT & SC-EDD], offer flexible and robust strategies that results in solutions with improved makespan and acceptable tardiness compared to single-rule approaches.

5.5. Research implication

Implications of the research findings for practical applications in production scheduling, particularly for industrial users are as follows. Firstly, the development of modified dispatching rules considering machine and job sequence-dependent setup times provides a valuable framework for optimizing production processes. Industrial users can implement these tailored rules to enhance their scheduling efficiency, leading to reduced idle times and improved machine utilization. By minimizing setup times and effectively managing job sequences, manufacturers can achieve higher throughput and better meet production targets.

Secondly, the application of multi-criteria decision-making analysis underscores the necessity for decision-makers to consider multiple performance measures when evaluating scheduling solutions. Industrial users often face competing system measures, such as minimizing makespan while reducing tardiness. This research enables a holistic approach to scheduling, allowing for the balancing of various performance metrics based on specific operational goals. Such flexibility can lead to improved customer satisfaction through timely deliveries and enhanced overall operational performance.

Thirdly, the findings emphasize the importance of adaptability in scheduling systems. As industries increasingly face demand fluctuations and changes in production requirements, the ability to adjust scheduling strategies becomes crucial. The proposed heuristics and dispatching rules can be easily modified to accommodate different production demand scenarios, enabling manufacturers to respond swiftly to market changes. This adaptability not only enhances operational resilience, but also positions companies to capitalize on emerging opportunities. Lastly, the research provides a foundation for further exploration of hybrid scheduling approaches that combine the strengths of various heuristics and optimization techniques. Industrial users can benefit from integrating advanced scheduling algorithms into their existing systems, leading to more sophisticated decision-making processes and improved production planning.

6. Conclusion

This study involved an unrelated parallel-machine scheduling problem with sequence- and machine-dependent setup times. Three system performance measures, including makespan, total tardiness, and the number of tardy jobs, are considered. For small problem instances, MILP models are formulated to find optimal solutions as well as Pareto solutions through the implementation of the AUGMECON method. For large problem instances, dispatching rule-based heuristics that consider sequence- and machine-dependent setup times are developed. They include SCT, SC-LPT, SC-EDD, and six combinations of these rules. For the combined rules, the appropriate rule-switching time is determined through computational experiments.

The heuristics are tested using 18 datasets of the monthly demand from the industry. The results indicate that under the high-demand scenario, the best dispatching rule for the makespan, total tardiness, and number of tardy jobs are SCT, SC-EDD, and [SCT & SC-EDD: 450], respectively. For the low-demand scenario, the best rule for makespan is [SCT & SC-EDD: 450], whereas [SC-EDD & SCT: 200] is most effective for both the total tardiness and the number of tardy jobs. In practice, the decision makers may assign multiple relative weights of importance to the performance measures, which determines the rule that should be used. We perform a complete enumeration of the weights to determine a series of conditions in which the DM prefers one rule over the others.

The result in this research is the stepping stone for several research directions in the future. One of them is to develop new dispatching rules or to extend the problem to multiple operations in sequence, where there are unrelated parallel machines in each operation. In addition, to further improve the performance, the dispatching rules can be implemented as a part of a meta-heuristic algorithm that generates solutions on the Pareto front for large problem instances.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.dajour.2024.100525.

Data availability

Data will be made available on request.

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